robic bacteria to several of the established chemotherapeutic agents, including the nitro-imidazoles.

Because of the relatively few patients and the low incidence of postoperative infection in this study, the clinical efficacy of tinidazole prophylaxis could not be firmly established. However, the suppressive effect of tinidazole on anaerobic bacteria in the female genital tract demonstrated in this study gives scientific support to the prophylactic use of nitro-imidazoles in gynaecological surgery.

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The Effect of Cyclical Administration of Levonorgestrel and Ethinyloestradiol on Blood Pressure, Body Mass, Blood Glucose and Serum Triglycerides

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SUMMARY

The effects of two differing dosage schedules of ethinyloestradiol and levonorgestrel, taken cyclically as oral contraceptives, on blood pressure, body mass, blood glucose and serum triglycerides were measured. After 12 months there were no significant changes in systolic and diastolic blood pressure or blood glucose levels. There was a small but statistically significant increase in body mass after 1 year in the group taking levonorgestrel 150 μg and ethinyloestradiol 30 μg ($P<0,02$), but not in the other group, although the mass in this group was significantly increased after 3 months of therapy. There was a statistically significant increase in serum triglycerides in the group taking levonorgestrel 50 μg with ethinyloestradiol 50 μg for 11 days, followed by levonorgestrel 125 μg with ethinyloestradiol 50 μg for 10 days. This effect appears to be dose-related and progressive with continued therapy: the significance of these results is discussed.

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Since oral contraceptives continue to be the most widely used steroid agents, the effects they have on basic physiological measurements and also on biochemical parameters are of paramount importance. A major change in oral contraceptive formulation in recent years has been the withdrawal of many products containing a daily dose of 50 μg oestrogen, following reports linking this dose level with a higher risk of circulatory and thromboembolic disease. Another major change has been the introduction of combination products containing a daily dose of only 30 μg oestrogen. This study was designed to study the effects on four parameters in patients taking an identical type of oestrogen and progestogen but in varying doses.

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SUBJECTS AND METHODS

Women of proven fertility, matched for age and parity, and attending the family planning clinics of the Department of Obstetrics and Gynaecology, University of the Witwatersrand, were randomly allocated to one of the two treatment groups.

Group A was given an oral contraceptive preparation containing 150 μg levonorgestrel and 30 μg ethinyloestradiol (Nordette; Wyeth) daily for 21 days starting on day 1 of the menstrual cycle. Placebo tablets were given on day 22 - 28.

Group B received a biphasic type of oral contraceptive (Biphasil, Wyeth; Normovlar ED; Schering) consisting of levonorgestrel 50 μg and ethinyloestradiol 50 μg for the first 11 days of the menstrual cycle, followed by levonorgestrel 125 μg and ethinyloestradiol 50 μg

for the following 10 days. Placebo tablets were given on day 22 - 28.

These products were chosen because the former contains the lowest dose of oestrogen (30 μg per day) in a combination oral contraceptive, whereas the biphasic preparation contains a higher oestrogen dose (50 μg per day), but the lowest total dose of hormone (oestrogen and progesterone) at present available in a combined oral contraceptive. This allowed a direct comparison between these dosage formulations.

Blood pressure, body mass, blood glucose values, and serum triglyceride levels were assessed before, after 3 months, and again after 1 year of therapy.

Women with any of the following were excluded from the trial: amenorrhoea; acute or chronic disease of the liver or gastro-intestinal tract; thrombophlebitis or a history of thrombo-embolism; carcinoma of the breast; overt gynaecological lesions; a history of hormone therapy, either by oral ingestion less than 3 months before the start of the trial, or by intramusclar injection less than 6 months before entering the trial.

RESULTS

Twenty-eight patients in group A and 26 patients in group B completed the trial.

The mean age of the patients in group A was 24,54 years (SD ± 4,01) and in group B 22,15 years (SD ± 5,21). The ages were statistically comparable.

The mean values for the various parameters are presented in Table I. The control values indicate that the two groups were comparable, and there was no significant difference in any of the parameters measured at 3 months when the two groups were compared.

However, after 12 months of therapy, although the other parameters show no significant difference, the mean serum triglyceride levels were significantly higher in group B, 119,6 mg/dl compared with 93,8 mg/dl ($P<0,05$).

A comparison of the changes occurring within each group for each of the parameters was also made by the paired t test. The results are presented in Table II.

In group A, the mean body mass increased significantly after 1 year of therapy, 55,3 kg compared with 56,3 kg ($P<0,02$). In group B, however, a significant increase was seen after 3 months, 57,1 kg compared with 57,9 kg ($P<0,05$), but after 1 year this effect was no longer significant.

The mean serum triglyceride levels were significantly increased in group B after both 3 months and 12 months, to 102,1 mg/dl at 3 months ($P<0,05$), and 119,6 mg/dl at 12 months ($P<0,001$).

No significant changes were recorded in blood glucose levels or blood pressure.

TABLE I. COMPARISON OF PARAMETERS MEASURED IN GROUPS A AND B

	Group	Control values			Values at 3 months			Values at 12 months		
		Mean	SD	SEM	Mean	SD	SEM	Mean	SD	SEM
Body mass (kg)	**A**	**55,32**	**9,64**	**1,82**	**55,57**	**9,92**	**1,87**	**56,32**	**10,26**	**1,94**
	B	**57,08**	**10,99**	**2,15**	**57,85**	**10,91**	**2,14**	**58,12**	**11,41**	**2,24**
Diastolic blood pressure	**A**	**76,71**	**7,94**	**1,50**	**73,57**	**7,80**	**1,47**	**72,86**	**9,66**	**1,83**
(mmHg)	**B**	**73,85**	**8,04**	**1,59**	**73,08**	**9,70**	**1,90**	**72,88**	**7,10**	**1,39**
Systolic blood pressure	**A**	**116,61**	**12,02**	**2,27**	**116,43**	**11,39**	**2,25**	**114,82**	**13,02**	**2,46**
(mmHg)	**B**	**115,00**	**8,94**	**1,75**	**114,62**	**13,49**	**2,64**	**117,12**	**10,60**	**2,08**
Blood glucose	**A**	**88,79**	**13,75**	**2,60**	**84,43**	**9,68**	**1,83**	**86,82**	**16,14**	**3,05**
(mg/dl)	**B**	**83,23**	**11,46**	**2,25**	**87,50**	**14,10**	**2,76**	**85,92**	**12,65**	**2,48**
Triglycerides	**A**	**93,00**	**35,93**	**6,79**	**92,36**	**31,02**	**5,86**	**93,79**	**38,25**	**7,23**
(mg/dl)	**B**	**88,96**	**33,76**	**6,62**	**102,08**	**40,63**	**7,97**	**119,62**	**51,34**	**10,07**

TABLE II. MEAN VALUES WITHIN EACH GROUP COMPARED BY THE PAIRED t TEST

	Group	Control value	Value at 3/12	Value at 12/12	t value		Degree of freedom		Significance	
					3/12	12/12	3/12	12/12	3/12	12/12
Body mass	**A**	**55,32**	**55,57**	**56,32**	**0,68**	**2,58**	**27**	**27**	**No**	**Yes†**
(kg)	**B**	**57,08**	**57,85**	**58,12**	**2,40**	**2,01**	**25**	**25**	**Yes***	**No**
Diastolic blood	**A**	**76,71**	**73,57**	**72,86**	**1,62**	**1,80**	**27**	**27**	**No**	**No**
pressure (mmHg)	**B**	**73,85**	**73,08**	**72,88**	**0,39**	**0,44**	**25**	**25**	**No**	**No**
Systolic blood	**A**	**116,61**	**116,43**	**114,82**	**0,55**	**0,86**	**27**	**27**	**No**	**No**
pressure (mmHg)	**B**	**115,00**	**114,62**	**117,12**	**0,17**	**0,90**	**25**	**25**	**No**	**No**
Blood glucose	**A**	**88,79**	**84,43**	**86,82**	**1,62**	**0,52**	**27**	**27**	**No**	**No**
(mg/dl)	**B**	**83,23**	**87,50**	**85,92**	**1,17**	**0,76**	**25**	**25**	**No**	**No**
Triglycerides	**A**	**93,00**	**92,36**	**93,79**	**0,14**	**0,14**	**27**	**27**	**No**	**No**
(mg/dl)	**B**	**88,96**	**102,08**	**119,62**	**2,10**	**4,27**	**25**	**25**	**Yes***	**Yes‡**

* $P<0,05$.
† $P<0,02$.
‡ $P<0,001$.

DISCUSSION

There have in the past been markedly varying reports on the effect of steroid hormones in oral contraceptives on carbohydrate metabolism, and the subject has been extensively reviewed.[1-4] That fasting blood glucose levels are unaffected is now generally accepted,[5-7] and the present study also shows that random blood glucose levels are not affected by this dosage schedule of steroid hormones after a year of use. A reasonable view in the light of present-day knowledge is that the disruption in glucose tolerance induced in some women by oestrogenic oral contraceptives is due to individual variation and is likely to be influenced by a diabetic predisposition and also by age and the type of oestrogen prescribed.

Blood pressure changes in women on steroid contraceptives have been recorded,[8,11] but the evidence is also overwhelmingly in favour of individual predisposition. This study confirms that in the vast majority of women low-dose combined oral contraceptives do not cause an elevation of either diastolic or systolic blood pressure.

Changes in body mass have been recorded in the past,[12,13] particularly when the earlier high-dose preparations were in use. Although a statistically significant increase in body mass was noted in group A after 1 year of treatment, the actual mean increase was small, only 1 kg. This does not take into account the cyclical variations in weight due to fluid retention, but overall the finding is reassuring as it indicates that long-term oral contraceptive ingestion is unlikely to cause significant weight gain, unless accompanied by an increased intake of kilojoules.

The changes in the serum triglyceride levels were highly significant and support a steadily increasing body of evidence indicating a link with circulatory disease and thrombo-embolism. It would also appear from this study that the increase in serum triglycerides is progressive, levels rising with continued ingestion. The study also confirms that the rise in serum triglyceride levels is related to the oestrogenic component of steroid contraceptives, the rise having been statistically significantly greater in group B (50 μg ethinyloestradiol per day) than in group A (30 μg ethinyloestradiol per day).

There is general agreement that a fasting hyperlipidaemia is consistently found in association with steroid hormone ingestion, and this is mainly due to an increase in triglyceride levels influenced by the oestrogenic component. However, combined oral contraceptives containing high doses of progestogen appear to induce greater changes than low-dose formulations containing the same amount of oestrogen.[14,15]

Plasma triglyceride kinetics have been investigated[16] in an attempt to clarify the situation, and it has been shown that both the pool size and the turnover rate of triglycerides are greatly increased by oestrogen alone or by combined preparations, but not by progesterone only.

Although not fully established, it would appear that this oestrogenic effect on triglycerides is a permanent biochemical change, and that triglyceride levels do not return to normal after long-term use in the majority of women.[17]

The mechanisms involved in this disturbance are probably complex, as the steady-state concentration in the fasting condition is obviously a balance between input and output of triglycerides from the plasma pool. These changes could be due to an increased input, a decreased output or both. At present the evidence available suggests that oestrogens increase triglyceride turnover, while progestogens increase triglyceride clearance. Plasma lipoprotein lipase is probably the enzyme involved and a rate-limiting factor. It is also possible that this change is induced by and secondary to a change in non-esterified fatty acids,[18-20] which appear to decrease significantly in women taking combined oral contraceptives.

More recently, the influence of oral contraceptives on high-density lipoprotein (HDL) cholesterol has been studied. The results have been conflicting, higher, lower and unchanged levels having been recorded.[21-23] It appears likely that oestrogens increase HDL cholesterol while progestins decrease it,[24] and the net effect of any given combination oral contraceptive will depend on its specific formulation. Because of the inverse relationship between coronary artery disease and HDL cholesterol level,[25,26] and the epidemiological evidence linking coronary artery disease and oral contraceptive ingestion, a re-examination of dosage schedules in use is indicated, since this may show that the risk of coronary artery disease varies from preparation to preparation.

The results of this study support the present-day move to low-dose oral contraceptive preparations; in particular, the oestrogenic component should be as low as is compatible with maximal protection from pregnancy. This trend will undoubtedly decrease the incidence of cardiovascular complications, thrombo-embolism and other, less major side-effects.

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